

AI Lead Generation & Outreach Bot

n8n Cloud Workflow — Technical Walkthrough

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Project: An end-to-end AI-powered lead generation and outreach automation pipeline built with n8n Cloud. The system reads leads from Google Sheets, generates personalized cold emails using Gemini AI, sends them via Gmail, and tracks all activity automatically with zero manual effort.

1 Workflow Architecture

The pipeline runs as a single n8n workflow triggered manually. Each node handles one responsibility in the outreach process.

Step	Node	Purpose
1	Manual Trigger	Starts the workflow on demand
2	Read Leads	Reads all rows from the Leads tab in Google Sheets
3	Filter Valid Leads	Removes rows with missing email or name fields
4	Generate Outreach Message	Basic LLM Chain v1.9 — sends lead data to Gemini AI for email generation
5	Prepare Outreach Data	Code node — pairs AI output with original lead data using \$() node reference
6	Save to Outreach Sheet	Appends or updates the Outreach tab with generated message and Pending status
7	Send Outreach Email	Sends the personalized email to the lead via Gmail OAuth2
8	Mark as Sent	Updates the Outreach sheet — status = Sent, sent_at = Pakistan timestamp

2 Architecture Decisions & Rationale

Manual Trigger Instead of Schedule

Lead outreach requires human review before sending. A manual trigger ensures the operator verifies the lead list before the pipeline runs, preventing accidental mass emails to unverified contacts.

Gemini AI for Personalization

Instead of a template-based approach, Gemini generates unique emails for each lead based on their role, industry, and specific pain point. This produces significantly higher response rates than generic outreach templates.

\$() Node Reference Fix for Data Pairing

n8n's Basic LLM Chain node drops input fields from its output — only the generated text passes through. The Prepare Outreach Data node solves this by using `$( "Filter Valid Leads" ).all()` to re-fetch the original lead data and pair it with the AI output by array index.

Upsert by Email in Outreach Sheet

Save to Outreach Sheet uses `appendOrUpdate` matched by email. This prevents duplicate rows when the pipeline is re-run and ensures the latest generated message always overwrites old data.

Name-Based Matching in Mark as Sent

Since multiple leads can share the same email in test data, Mark as Sent matches by name for more precise row targeting. In production, email matching would be preferred.

Pakistan Timezone for Timestamps

All `sent_at` timestamps use Asia/Karachi timezone via Luxon's `setZone()` method, ensuring timestamps are meaningful for local business tracking.

3 Assumptions Made

Lead Data

- Leads are manually added to the Google Sheets Leads tab before running the workflow.
- Each lead has a valid email, name, company, role, industry, and `pain_point` field.
- Pain points are specific enough for Gemini to generate meaningful personalized content.

Email Deliverability

- Gmail OAuth2 credentials remain valid — no token expiry during pipeline runs.
- Lead emails are real and deliverable — no email validation is performed by the workflow.
- Gmail sending limits are not exceeded for the volume of leads being processed.

AI Output

- Gemini returns clean email body text without subject lines or extra formatting.
- Generated emails are professional and appropriate — no content moderation layer is applied.
- Gemini API daily limit is sufficient for the number of leads in each batch.

4 Error Handling & Edge Cases

Scenario	How It Is Handled
Missing email or name	Filter Valid Leads removes any row with empty email before processing
LLM Chain drops input fields	Prepare Outreach Data uses <code>\$(“Filter Valid Leads”).all()</code> to re-fetch original lead data by index
Lead already contacted	Filter checks Outreach sheet for existing Sent status — skips already contacted leads
Gemini API rate limit	Failed items are skipped — re-run the workflow to process remaining leads
Gmail sending failure	n8n stops workflow on error — operator is notified and can retry manually
Duplicate leads in sheet	<code>appendOrUpdate</code> operation by email prevents duplicate rows in Outreach sheet
Wrong timestamp timezone	All timestamps use Asia/Karachi timezone explicitly via <code>Luxon setZone()</code>

5 Known Limitations & Future Improvements

Current Limitations

- **No webhook trigger:** Pipeline must be run manually — no real-time lead processing.
- **No email validation:** Fake or invalid emails cause Gmail to fail silently.
- **Gemini free tier:** 20 API calls/day limits batch size for production use.
- **No follow-up automation:** Only one email per lead — no drip sequence.
- **Name-based matching:** Duplicate names in sheet could cause wrong row updates.

What I Would Improve With More Time

- **Automated lead scraping:** Integrate Apollo.io or LinkedIn API to auto-populate leads.
- **Follow-up sequence:** Schedule 3-touch email drip if no reply after 3 days.
- **Reply detection:** Gmail webhook to detect replies and stop follow-ups automatically.
- **A/B testing:** Generate 2 email variants per lead and track which performs better.
- **Slack notifications:** Alert when a lead replies or when batch completes.

6 Technology Stack

Tool	Usage	Reason
n8n Cloud	Workflow orchestration	Visual builder with code nodes — perfect for hybrid automation pipelines
Google Gemini AI	Personalized email generation	Strong language understanding, free tier available, JSON-friendly output
Google Sheets	Lead storage and CRM tracking	Easy to manage leads manually, native integration with n8n
Gmail OAuth2	Email dispatch	Reliable delivery, OAuth2 security, supports custom sender name

JavaScript (Code node)	Data pairing and transformation	Needed to fix LLM Chain data drop issue using \$() node reference
Luxon (built-in n8n)	Timezone handling	\$now.setZone() for accurate Pakistan time in sent_at timestamps

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